ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 600 Pa	240 / 0.4 = 600.
2	**(d)** more air is pumped in	More collisions.
3	**(c)** the same type of storm with different names	Tropical cyclones.
4	**(b)** 500 Pa	20/0.04 = 500 Pa.
5	**(b)** outside pressure is lower than inside	Lower external pressure.
6	**(b)** over land under thunderclouds	From severe thunderstorms.
7	**(b)** 200 Pa	20/0.1 = 200 Pa.
8	**(d)** land is warmer than sea in the daytime	Air rises over land.
9	**(b)** eye wall	Around the eye.
10	**(a)** 30000 Pa	600/0.02 = 30000 Pa.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	P = 80/0.2 (1); = 400 Pa (1).	1+1
12	Heated air expands and becomes less dense (1); The balloon floats up (1).	1+1
13	Air tries to equalise the pressure difference (1); Moves from where the pressure is more (1).	1+1
14	Many nails give a large total area (1); Pressure on the body is small (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	P = 400 / 0.02 = 20000 Pa (1); Two feet: area 0.04, P = 10000 Pa (1); Pressure is double on one foot (1).	1+1+1
16	Cyclone: large, spirals over warm sea (1); Tornado: small, funnel on land (1); Thunderstorm: lightning and rain (1).	1+1+1
17	Squeezing the bulb removes air (1); Releasing it lowers the pressure inside (1); Atmospheric pressure fills it with liquid (1).	1+1+1
	OR: Bottle with holes at heights (1); Lowest hole shows the farthest jet (1); Pressure increases with depth (1).	1+1+1
18	Green: no action (1); Yellow/orange: be aware, be prepared (1); Red: take action (1).	1+1+1
	OR: Take lightning discharge to ground (1); Protect building and people (1); Pointed metal rod (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tractor area 1.2 square metres (1); Pressure 36000/1.2 = 30000 Pa (1); Person: 600/0.05 = 12000 Pa (1); Tractor pressure is greater but wheels are wide (1); Wide tyres prevent sinking (1).	1+1+1+1+1
	OR: Before: warnings, shelters, supplies (1); Before: secure loose objects (1); During: stay inside (1); After: avoid flood water and wires (1); Help and report (1).	1+1+1+1+1
20	Pressure rises with depth (1); Acts in all directions (1); Jets from side holes (1); Dam wall is thicker at the base (1); Independent of container shape (1).	1+1+1+1+1
	OR: Faster particles collide more (1); More force on walls (1); Cooker traps steam (1); Boiling point rises (1); Food cooks faster (1).	1+1+1+1+1

Mistakes to avoid:

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** rising	Warm air rises.
2	**(c)** air sinks at the centre	Sinking air.
3	**(a)** doubles	The area of contact is halved while the weight is unchanged.
4	**(c)** the sea to the land in summer	Bring rain.
5	**(d)** warm sea, rising air, low pressure, inward spiral	Process.
6	**(d)** 20 square metres	A = F/P = 500/25.
7	**(a)** it loses the supply of warm moist air	Energy source is lost.
8	**(c)** save lives by timely evacuation	Preparedness.
9	**(b)** 3000 N	F = P x A.
10	**(a)** air leaks in so pressure equalises	Seal is lost.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	20/4 (1); = 5 N per square cm (1).	1+1
12	Sea becomes dangerous (1); They can return to shore in time (1).	1+1
13	Air pressure changes with height (1); Pressure inside the ear adjusts (1).	1+1
14	Steam builds up pressure (1); Pressure pushes the lid up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest face 0.02 sq m: 30/0.02 = 1500 Pa (1); Smallest face 0.005 sq m: 30/0.005 = 6000 Pa (1); Smaller area gives greater pressure (1).	1+1+1
16	Land warms faster than sea (1); Warm air rises over land (1); Cool sea air flows in (1).	1+1+1
17	More weight of water above (1); Pressure is higher at depth (1); Base made thicker (1).	1+1+1
	OR: Sucking lowers air pressure in the straw (1); Outside atmospheric pressure acts on liquid (1); Liquid is pushed up (1).	1+1+1
18	Atmospheric pressure is lower (1); Water boils at a lower temperature (1); Cooking takes longer (1).	1+1+1
	OR: Wide tyres spread weight and reduce pressure (1); Treads increase grip (1); Sports shoes use spikes for higher pressure on ground (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day: land warmer, air rises (1); Low pressure over land (1); Sea breeze (1); Night: sea warmer, land breeze (1); Moderates coastal temperature (1).	1+1+1+1+1
	OR: Uneven heating makes pressure differences (1); Air moves from high to low (1); Rotation turns winds in spirals (1); Cyclones spin (1); Monsoons result from seasonal differences (1).	1+1+1+1+1
20	Warm sea, rising moist air (1); Low pressure (1); Eye: calm (1); Eye wall: strongest winds (1); Effects: storm surge, rain, wind damage (1).	1+1+1+1+1
	OR: Dams: thick base (1); Vehicles: wide tyres (1); Balloons: hot air (1); Pumps: lower pressure (1); Cookers: raised pressure (1).	1+1+1+1+1

Mistakes to avoid:

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- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.

- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** reduce the force of storm surges	Natural barrier.
2	**(c)** increases	Depth increases.
3	**(d)** atmospheric pressure pushes the liquid	Low pressure inside.
4	**(c)** lightning, thunder and heavy rain	Typical.
5	**(d)** equal in all directions	Equal.
6	**(d)** the liquid with the greatest density	Pressure depends on density and depth.
7	**(a)** a breeze	Light wind.
8	**(b)** depth of water	Depth only.
9	**(c)** squeezing and releasing the bulb to change air pressure	Air pressure.
10	**(b)** extreme danger, take action	Highest warning.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wind provides lift (1); Moving air creates pressure difference (1).	1+1
12	More liquid above the bottom (1); Pressure increases with depth (1).	1+1
13	To avoid storm-surge flood water (1); Safe from damage (1).	1+1
14	Tall objects attract lightning (1); Charge can pass through the tree (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heated can with a little water is sealed and cooled (1); Steam condenses, pressure inside falls (1); Outside pressure crushes the can (1).	1+1+1
16	Gas pressure inside increases (1); Walls cannot withstand (1); Bursts (1).	1+1+1
17	Warm water gives moisture and energy (1); Evaporation fuels the storm (1); Above about 26 degrees Celsius (1).	1+1+1
	OR: Pressure outside is low at high altitude (1); Passengers need a comfortable pressure (1); Prevents breathing and ear problems (1).	1+1+1
18	Cyclone shelters (1); Strong buildings (1); Plant trees and mangroves (1).	1+1+1
	OR: Storm surge floods the coast (1); Strong winds (1); Heavy rain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Straw sipping (1); Suction cup (1); Card on an inverted water-filled glass (1); Collapsing can (1); Pressure acts in all directions (1).	1+1+1+1+1
	OR: Government: evacuation, shelters (1); Family: supplies, safe room (1); Fishermen: return to harbour (1); Warnings through radio (1); Follow official advice (1).	1+1+1+1+1
20	Pressure is greater in the 6 m tank (1); It is about double (1); Jet is faster and goes farther (1); Pressure depends on depth (1); Not on tank width (1).	1+1+1+1+1
	OR: Charges separate in clouds (1); Large discharge to ground (1); Thunder is the sound (1); Conductor gives an easy path to earth (1); Protects buildings (1).	1+1+1+1+1

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MASTERMIND